

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

2D Steady Diffusion-Convection Analysis with FVM

NUMERICAL FLOW SIMULATION - ASSESSMENT 1

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0 Introduction

We want to study the temperature field solution $T(x, y)$ of the steady heat equation,

$$\nabla \cdot (\rho \mathbf{u} T) = \nabla \cdot (\Gamma \nabla T) \tag{1}$$

We consider a fluid flowing through a plane channel between horizontal walls located in $y = 0$ and $y = H$, where the density ρ, velocity field u and thermal properties are known ($\Gamma = k/c_p$ is the ratio of thermal conductivity k to specific heat capacity c_p). We assume that T is a passive scalar, i.e. the fluid properties and the velocity field are independent of temperature. The two-dimensional numerical domain is a simple rectangle such that $0 \leq x \leq L$ and $0 \leq y \leq H$.

1 Expression of the velocity field as function of the mean and maximum streamwise velocities

To compute the velocity field $\mathbf{u} = (u_x, u_y)$, we start from the 2D steady Navier-Stokes equations for incompressible fluids, obtained assuming ρ and μ to be constant:

$$\begin{cases} \dfrac{\partial u_x}{\partial x} + \dfrac{\partial u_y}{\partial y} = 0 \\ \rho u_x \dfrac{\partial u_x}{\partial x} + \rho u_y \dfrac{\partial u_x}{\partial y} = -\dfrac{\partial p}{\partial x} + \mu \dfrac{\partial^2 u_x}{\partial x^2} + \mu \dfrac{\partial^2 u_x}{\partial y^2} \\ \rho u_x \dfrac{\partial u_y}{\partial x} + \rho u_y \dfrac{\partial u_y}{\partial y} = -\dfrac{\partial p}{\partial y} + \mu \dfrac{\partial^2 u_y}{\partial x^2} + \mu \dfrac{\partial^2 u_y}{\partial y^2} \end{cases}$$

Since $\mathbf{u} = \mathbf{u}(y)$ (fully developed flow), we can conclude that:

$$\frac{\partial u_x}{\partial x} = \frac{\partial u_y}{\partial x} = 0$$

Hence, by the mass conservation equation, we get:

$$\frac{\partial u_y}{\partial y} = 0$$

This means that both the partial derivatives of u_y are null and therefore:

$$u_y \equiv \text{const}$$

However, by the no slip condition on the walls, we have $u_y(0) = u_y(H) = 0$ and so:

$$u_y \equiv 0$$

At this point, the momentum equation along y simply reduces to:

$$\frac{\partial p}{\partial y} = 0 \Rightarrow p = p(x)$$

Now, let us focus on the first momentum equation. Using the results we just obtained, it can be simplified as follows:

$$\mu \frac{\partial^2 u_x}{\partial y^2} = \frac{\partial p}{\partial x}$$

We observe that the left-hand side depends only on y, whereas the right-hand side depends only on x. The only way this equality can hold is if both sides are constant:

$$\mu \frac{\partial^2 u_x}{\partial y^2} = \frac{\partial p}{\partial x} = const$$

At this stage, we can integrate both sides twice to obtain the expression of $u_x(y)$:

$$\int \frac{\partial^2 u_x}{\partial y^2} dy = \int \frac{1}{\mu} \frac{\partial p}{\partial x} dy \quad \Rightarrow \quad \frac{\partial u_x}{\partial y} = \frac{y}{\mu} \frac{\partial p}{\partial x} + C_1$$

$$\int \frac{\partial u_x}{\partial y} dy = \int \left(\frac{y}{\mu} \frac{\partial p}{\partial x} + C_1 \right) dy \quad \Rightarrow \quad u_x(y) = \frac{y^2}{2\mu} \frac{\partial p}{\partial x} + C_1 y + C_2$$

To find the values of the two integration constants, we impose the boundary conditions (no slip condition on the walls):

$$u_x(0) = 0 \quad \Rightarrow \quad C_2 = 0$$
$$u_x(H) = 0 \quad \Rightarrow \quad C_1 = -\frac{H}{2\mu} \frac{\partial p}{\partial x}$$

Hence,

$$u_x(y) = -\frac{1}{2\mu} \frac{\partial p}{\partial x} y(H - y)$$

According to the given definition, we compute u_{mean}

$$u_{mean} = \frac{1}{H} \int_0^H u_x(y) dy = \frac{1}{H} \int_0^H -\frac{1}{2\mu} \frac{\partial p}{\partial x} y(H - y) dy = -\frac{H^2}{12\mu} \frac{\partial p}{\partial x}$$

Manipulating $u_x(y)$ with basic algebra, we are able to find the requested expression as a function of u_{mean}:

$$u_x(y) = -\frac{6}{12} \frac{H^2}{H^2} \frac{\partial p}{\partial x} y(H - y) = \frac{6u_{mean}}{H} y \left(1 - \frac{y}{H}\right)$$

Therefore, the velocity field as a function of the mean velocity takes the following form:

$$\boxed{\mathbf{u}(y) = \left(\frac{6u_{mean}}{H} y \left(1 - \frac{y}{H}\right), 0 \right)}$$

Given that the horizontal velocity profile is parabolic, the maximum streamwise velocity can be found at the midpoint between the two walls:

$$u_{max} = u_x \left(\frac{H}{2} \right) = -\frac{1}{8\mu} \frac{\partial p}{\partial x} H^2$$

We get:

$$u_x(y) = -\frac{4}{8} \frac{H^2}{H^2} \frac{\partial p}{\partial x} y(H - y) = \frac{4u_{max}}{H} y \left(1 - \frac{y}{H}\right)$$

As a result, the velocity field as a function of the maximum streamwise velocity reads as follows:

$$\boxed{\mathbf{u}(y) = \left(\frac{4u_{max}}{H} y \left(1 - \frac{y}{H}\right), 0 \right)}$$

2 Integration of the heat equation on internal control volumes (CVs)

To numerically solve the steady heat equation in a 2D domain, we create a uniform structured mesh, assembling control volumes as the one shown in figure 1, where P is the node associated with it, central with respect to its CV, and N, S, W, E are the nodes associated with the CV neighbor control volumes. We also indicate n, s, w, e as CV faces. The dimensions of the control volume are Δx in width and Δy in height. The key is to find the steady equation in each control volume, simplify it thanks to the simplicity of the domain and the appropriate numerical discretizations, and assemble a linear system composed of all control volumes' equations. We start by integrating (1) in a single CV.

$$\int_{V_{CV}} \nabla \cdot (\rho \mathbf{u} T) \, dV = \int_{V_{CV}} \nabla \cdot (\Gamma \nabla T) \, dV \tag{2}$$

Applying the divergence theorem $\int_V \nabla \cdot \mathbf{F} \, dV = \oint_S \mathbf{F} \cdot \mathbf{n} \, dS$, we obtain the following.

$$\oint_{S_{CV}} \rho T \mathbf{u} \cdot \mathbf{n} \, dS = \oint_{S_{CV}} \Gamma \nabla T \cdot \mathbf{n} \, dS \tag{3}$$

We develop the integral of the surface on CV faces and the scalar product between the velocity field $\mathbf{u}$ and the unit normal vectors $\mathbf{n}_e = (1, 0)$, $\mathbf{n}_w = (-1, 0)$, $\mathbf{n}_n = (0, 1)$, $\mathbf{n}_s = (0, -1)$ for each face. We recall that, as shown in the previous point, $\mathbf{u} = (u_x, 0)$. Thus, we obtain the following:

$$\int_e \rho T u_x \, dy - \int_w \rho T u_x \, dy + \cancel{\int_n \rho T u_y \, dx} - \cancel{\int_s \rho T u_y \, dx} = \int_e \Gamma \frac{\partial T}{\partial x} \, dy - \int_w \Gamma \frac{\partial T}{\partial x} \, dy + \int_n \Gamma \frac{\partial T}{\partial y} \, dx - \int_s \Gamma \frac{\partial T}{\partial y} \, dx$$

Finally, we consider a reasonable approximation to treat the integral arguments as constants, with respect to the integral measure. Therefore, we obtain the following:

$$(\rho u_x)_e T_e \Delta y - (\rho u_x)_w T_w \Delta y = \Delta y \left(\Gamma \frac{\partial T}{\partial x} \right)_e - \Delta y \left(\Gamma \frac{\partial T}{\partial x} \right)_w + \Delta x \left(\Gamma \frac{\partial T}{\partial y} \right)_n - \Delta x \left(\Gamma \frac{\partial T}{\partial y} \right)_s \tag{4}$$

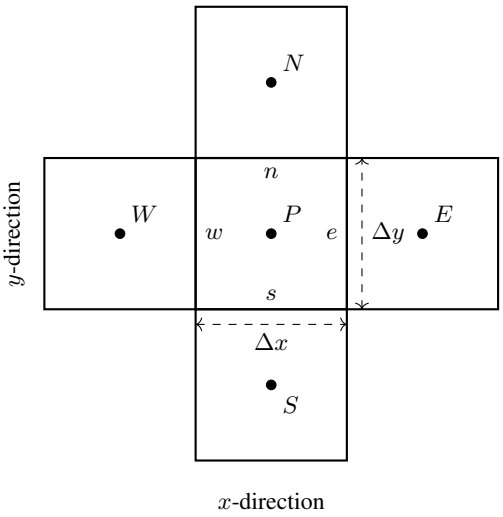


FIGURE 1
Internal control volume.

3 Numerical discretization

We now discretize the previously obtained equation.

- We discretize the diffusion term using central differencing (CD).

$$T_e \approx \frac{T_P + T_E}{2} \, , \left(\frac{\partial T}{\partial x}\right)_e \approx \frac{T_E - T_P}{\delta x_{PE}}$$

$$T_w \approx \frac{T_W + T_P}{2} \, , \left(\frac{\partial T}{\partial x}\right)_w \approx \frac{T_P - T_W}{\delta x_{WP}}$$

$$T_s \approx \frac{T_P + T_S}{2} \, , \left(\frac{\partial T}{\partial y}\right)_s \approx \frac{T_P - T_S}{\delta y_{SP}}$$

$$T_n \approx \frac{T_P + T_N}{2} \, , \left(\frac{\partial T}{\partial y}\right)_n \approx \frac{T_N - T_P}{\delta y_{PN}}$$

- We discretize the convective term using upward differencing (UD), with $(\rho u_x)_e > 0$ and $(\rho u_x)_w > 0$.

$$T_e \approx T_P \, , T_w \approx T_W$$

4 Algebraic equation and dimensionality check

Substituting the previous discretizations in (4), the equation takes the following form:

$$(\rho u_x)_e T_P \Delta y - (\rho u_x)_w T_W \Delta y = \Gamma_e \frac{T_E - T_P}{\delta x_{PE}} \Delta y - \Gamma_w \frac{T_P - T_W}{\delta x_{WP}} \Delta y + \Gamma_n \frac{T_N - T_P}{\delta y_{PN}} \Delta x - \Gamma_s \frac{T_P - T_S}{\delta y_{SP}} \Delta x \quad (5)$$

To simplify the notation, we introduce the following quantities:

$$D_e = \frac{\Gamma_e}{\delta x_{PE}}, \quad D_w = \frac{\Gamma_w}{\delta x_{WP}}, \quad D_n = \frac{\Gamma_n}{\delta y_{PN}}, \quad D_s = \frac{\Gamma_s}{\delta y_{SP}}$$

$$F_e(y_P) = \rho_e u_x(y_P) \Delta y, \qquad F_w(y_P) = \rho_w u_x(y_P) \Delta y$$

where y_P is the y-coordinate of node P

At this stage, the algebraic equation can be written as follows:

$$a_P T_P = a_E T_E + a_W T_W + a_N T_N + a_S T_S \quad (6)$$

where:

- $a_P = F_e + (D_e + D_w)\Delta y + (D_n + D_s)\Delta x$
- $a_E = D_e \Delta y$
- $a_W = F_w + D_w \Delta y$
- $a_N = D_n \Delta x$
- $a_S = D_s \Delta x$

For what concerns the dimensionality check, we need to verify that the convective terms $F_\cdot$ have the same unit measure as the diffusion terms $D_\cdot \Delta \cdot$
(Caveat: M refers to mass, L refers to length, T refers to Time, Θ refers to Temperature)

$$[F_\cdot] = [\rho_\cdot u_x \Delta \cdot] = \frac{M}{L^3}\frac{L}{T}L = \frac{M}{LT}$$

$$[D_\cdot \Delta \cdot] = \left[\frac{\Gamma_\cdot}{\delta_\cdot}\Delta \cdot\right] = \frac{\frac{M}{LT}}{L}L = \frac{M}{LT}$$

where we found the unit of measure of Γ as follows: $[\Gamma] = \left[\frac{k}{c_P}\right] = \frac{\frac{ML}{T^3\Theta}}{\frac{ML^2}{\Theta MT^2}} = \frac{M}{LT}$

5 Control volumes indexing

Considering a structured mesh composed of control volumes, we need a convention to identify them in the structure through indexing. We chose the convention commonly used to identify matrix indices. For example, the third CV in the third row, counting from the reader's left to right horizontally, supposing that there are more than three CVs in a row, has indices $i = 3$, $j = 3$. In order to clarify, we provide a sketch in figure 2.

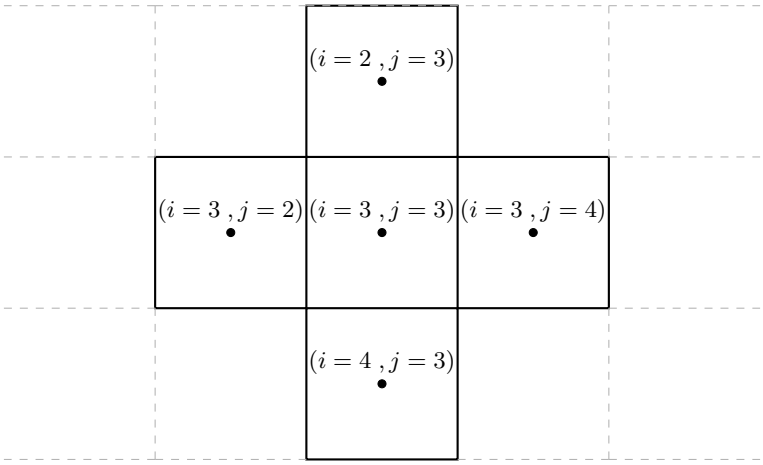


FIGURE 2
Control volumes indexing - fragment of the structured mesh.

In particular, the indices of control volumes touching the four boundaries are the following:

- **Top wall indices** : $(1, j)$, $\forall j = 1, ..., n_x$.
- **Bottom wall indices** : (n_y, j) , $\forall j = 1, ..., n_x$.
- **Left (inlet) indices** : $(i, 1)$, $\forall i = 1, ..., n_y$.
- **Right (outlet) indices** : (i, n_x) , $\forall i = 1, ..., n_y$.

We specify that n_x is the number of nodes in the x-direction, thus the number of columns, and n_y is the number of nodes in the y-direction, thus the number of rows.
Finally, it is important to underline the indices of the neighbor control volumes, given a control volume with indices $(i\ , j)$. Taking into account the previous notation with the N, S, W, E neighbor nodes associated with the respective control volumes, we have the following indexing:

- **N neighbor** : $(i - 1\ , j)$.

- **S neighbor** : $(i+1\ , j)$.
- **W neighbor** : $(i\ , j-1)$.
- **E neighbor** : $(i\ , j+1)$.

6 Degrees of freedom indexing

For the DOFs indexing, we chose to start from the top-left corner and proceed horizontally, as shown in figure 3.

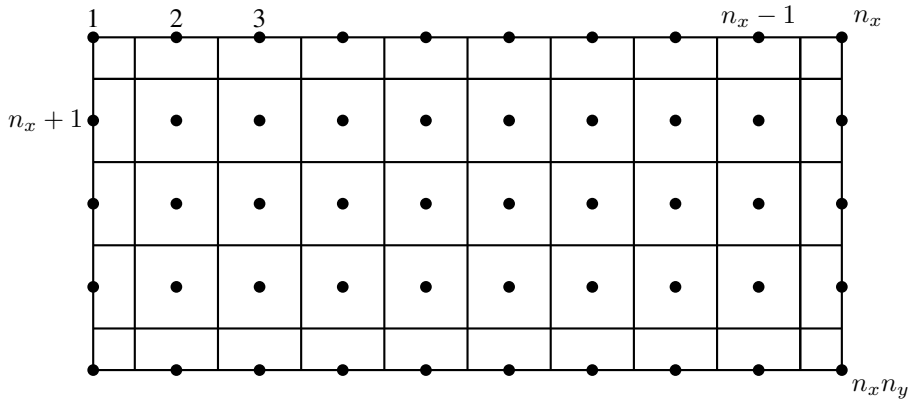


FIGURE 3
Indices of the degrees of freedom

The relationship between the DOFs indexing n and the row and column indices of the mesh i and j can be expressed as:

$$n = (i-1)n_x + j$$

The boundary DOFs happen to have the following indices:

- **inlet**: $n = 1 + n_x(i-1), \quad$ for $i = 1, ..., n_y$
- **outlet**: $n = n_x i, \quad$ for $i = 1, ..., n_y$
- **top wall**: $n = j, \quad$ for $j = 1, ..., n_x$
- **bottom wall**: $n = j + n_x(n_y - 1)$ for $j = 1, ..., n_x$

The indices of the neighbors of a given CV of DOF index n are shown in figure 4.

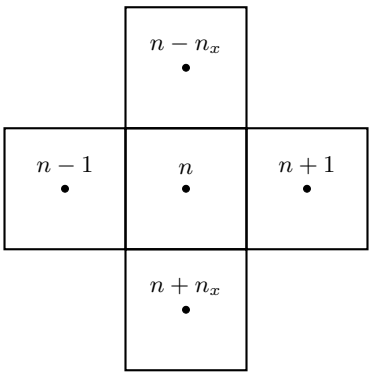


FIGURE 4
Neighbors indices

To define the matrix $\mathbf{A}$ and the vector $\mathbf{b}$, we used the following snippet of MATLAB code:

```
y = linspace(0, H, ny);
u_y = (6 * u_mean / H^3) .* y .* (H - y);
F_e = rho * u_y * dy;
F_e = repmat(F_e,nx,1);
F_w = F_e;
%% INNER CVs
```

```
% We start by just defining the matrix and the known vector b for
% inner CVs, ignoring the BCs
a_p = (D_e + D_w)*dy + (D_n + D_s)*dx;
a_n = D_n * dx;
a_s = D_s * dx;
a_w = D_w*dy; %UW for convection with positive velocity field
a_e = D_e*dy;
e = ones(N,1);
p_diag = e*a_p + F_e;
w_diag = -e*a_w - F_w;
n_diag = -e*a_n;
s_diag = -e*a_s;
e_diag = -e*a_e;
b_val = 0;
A_sparse = spdiags([n_diag w_diag p_diag e_diag s_diag], [-nx -1 0 +1 +nx], N, N);
A = full(A_sparse);
b = e*b_val;
```

LISTING 1
Assembly of A and b for inner CVs

7 Inlet boundary conditions

Taking into account the inlet column nodes located exactly along the inlet boundary, each of them must satisfy the following discrete equation.

$$T_P = 0T_E + 0T_W + 0T_S + 0T_N + T_{in}$$

Thus,

$$T_P = T_{in}$$

where P is a node in the inlet column and $N,\ S,\ W,\ E$ are its neighbor nodes.
We assume that nodes associated with top and bottom corner CVs belong, respectively, to upper wall and lower wall. We developed the previous boundary condition in MATLAB as shown in the code below.

```
%% INLET (left) Dirichlet BCs
% We consider the CV nodes as positioned on the bound, with the CVs having
% half the width
left_indices = (nx + 1) : nx : ((ny-2)*nx + 1);
for ii = left_indices
    A(ii,:) = 0;
    A(ii,ii) = 1;
    b(ii) = T_in;
end
```

LISTING 2
Implementation of Dirichlet BCs at the inlet

8 Outlet boundary conditions

For the implementation of Neumann boundary conditions, we again use an upwind scheme for the convective terms and central differencing for the diffusive ones. We remark that, when integrating along the x-axis, a factor of one half appears. Indeed, the outlet CVs have half the width as the internal ones. Moreover, the east diffusive term disappears because of the homogeneous Neumann boundary conditions. As a result, we get the following equation:

$$F_eT_P - F_wT_W = -D_w(T_P - T_W)\Delta y + D_n(T_N - T_P)\frac{\Delta x}{2} - D_s(T_P - T_S)\frac{\Delta x}{2}$$

$$\Rightarrow a_PT_P = a_ET_E + a_WT_W + a_NT_N + a_ST_S$$

where:

- $a_P = F_e + (D_n + D_s)\dfrac{\Delta x}{2} + D_w\Delta y$
- $a_E = 0$
- $a_W = F_w + D_w\Delta y$
- $a_N = D_n\dfrac{\Delta x}{2}$
- $a_S = D_s\dfrac{\Delta x}{2}$

To implement these boundary conditions in MATLAB, we used the following lines:

```
%% OUTLET (right) Neumann BCs
% Nodes defined as before
right_indices = 2*nx : nx : nx*(ny-1);
for jj = right_indices
    A(jj,:) = 0;
    A(jj,jj) = F_e(jj) + (D_n + D_s)*(dx/2) + D_w*dy;
    A(jj,jj-1) = - F_w(jj) - D_w*dy;
    if jj-nx > 0
        A(jj,jj-nx) = - D_n * dx / 2;
    end
    if jj+nx <= N
        A(jj,jj+nx) = - D_s * dx / 2;
    end
end
```

LISTING 3
Implementation of Neumann BCs on the outlet

9 Walls boundary conditions

Given a Dirichlet boundary condition on the walls $T = T_{wall}$ and considering the nodes in the first and last rows of CVs located exactly on the upper and lower walls, we impose that each of them must satisfy the following discrete equation.

$$T_P = 0T_E + 0T_W + 0T_S + 0T_N + T_{wall}$$

Thus,

$$T_P = T_{wall}$$

where P is a node in the upper wall row or in the lower wall row and $N,\ S,\ W,\ E$ are its neighbor nodes.
We developed the previous boundary condition in MATLAB as shown in the code below.

```
%% TOP and BOTTOM (walls) BCs
% We assume that the corners belong to the walls
top_indices = 1:nx;
for ii = top_indices
    A(ii,:) = 0;
    A(ii,ii) = 1;
    b(ii) = T_wall;
end
bottom_indices = (N-nx+1) : N;
for ii = bottom_indices
    A(ii,:) = 0;
    A(ii,ii) = 1;
    b(ii) = T_wall;
end
```

LISTING 4
Implementation of Dirichlet BCs on the walls

10 Calculation of the mean streamwise velocity and the Reynolds number

First, let us compute the value of Γ:

$$\Gamma = \frac{k}{c_p} = \frac{0.12\,\mathrm{W\,m^{-1}\,K^{-1}}}{10\,\mathrm{J\,kg^{-1}\,K^{-1}}} = 0.012\,\mathrm{kg\,m^{-1}\,s^{-1}}$$

Starting from the expression of the Péclet number, we can easily compute the mean streamwise velocity:

$$u_{\text{mean}} = \frac{\Gamma\, Pe}{2H\rho} = \frac{0.012\,\mathrm{kg\,m^{-1}\,s^{-1}} \cdot 16.5}{2 \cdot 1\,\mathrm{m} \cdot 1\,\mathrm{kg/m^3}} = 0.099\,\mathrm{m\,s^{-1}}$$

Then, we can compute the Reynolds number as:

$$Re = \frac{2H u_{\text{mean}}}{\nu} = \frac{2 \cdot 1\,\mathrm{m} \cdot 0.099\,\mathrm{m\,s^{-1}}}{0.01\,\mathrm{m^2\,s^{-1}}} = 19.8$$

Since $19.8 \ll 1000$, which is the critical threshold used in the literature for plane channels, we can actually assume a laminar flow.

11 Linear system solution (with backslash)

We assembled the matrix A and the right-hand side b as shown above. At this point, we defined the problem data and generated the discretization grid.

```
rho = 1; % mass density [kg/m^3]
c_p = 10; % specific heat capacity [J/(K*kg)]
k = 0.12; % thermal conductivity [W/(m*K)]
nu = 1e-2; % kinematic viscosity [m^2/s]
gamma_ = k / c_p; % diffusivity per unit density
L = 10; % Channel length [m]
H = 1; % Channel height [m]
Pe = 16.5; % Peclet number
u_mean = (Pe * gamma_) / (rho * 2*H); % Mean velocity
Re = 2*H * u_mean / nu; % Reynolds number
T_in = 50; % Inlet temperature [C]
T_wall = 100; % Wall temperature [C]
```

LISTING 5
Problem data

```
nx = 50; % number of columns of the grid
ny = 5; % number of rows of the grid
dx = L / (nx - 1);
dy = H / (ny - 1);
N = nx * ny; %DOFs
y = linspace(0, H, ny);
x = linspace(0, L, nx);
```

LISTING 6
Grid generation

Then we solved the linear system with backslash $T = A \backslash b$ and plotted the surf (figure 5) and contour (figure 7) of the solution to check if it was physically acceptable.

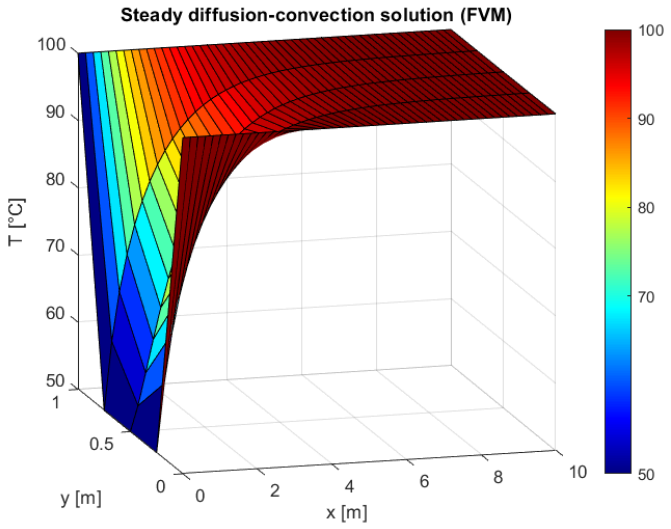


FIGURE 5
Temperature Surf Plot

We notice that the temperature increases along the x direction, reaching a maximum close to T_{wall} at the outlet, due to the diffusion effect from the walls. Near the inlet, the convection effect keeps the temperature lower because of velocity field flowing and transporting the colder fluid from the inlet through the vane. As physically expected, the temperature in the middle of the vane, i.e. $T(x, H/2)$ is the lowest $\forall x \in [0, L]$.

12 Boundary conditions plots

To demonstrate *a posteriori* that the boundary conditions have been well-imposed, we show the following four plots. Three of them illustrate the temperature profile at the inlet and at the two walls. Note that in the inlet plot the temperature is equal to T_{wall} at the two extrema. This is because we assumed these two DOFs to belong to the walls. Moreover, to show the homogeneous Neumann boundary condition at the outlet, we computed the difference between the temperature in the last column and the temperature in the penultimate column and we divided it by the horizontal grid size. The fourth plot shows that this difference is almost zero, as expected.

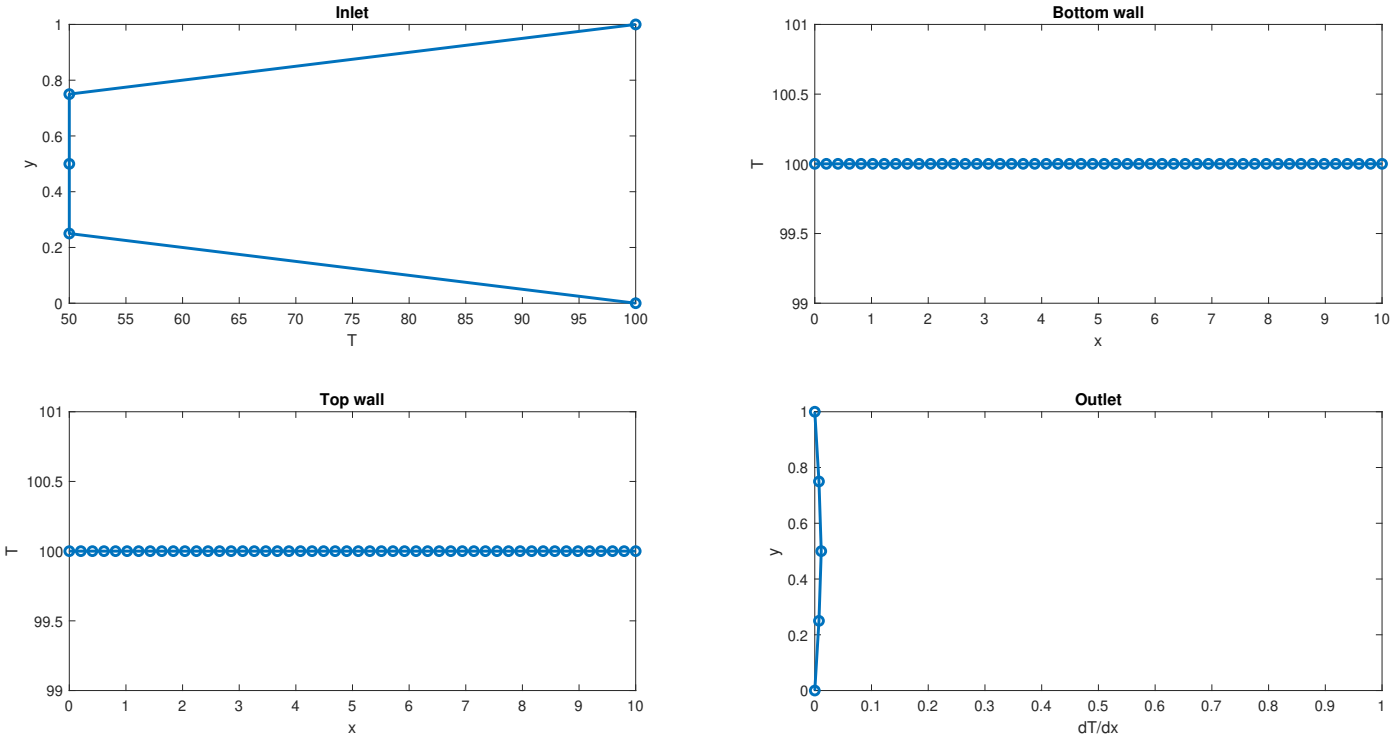


FIGURE 6
Plots of the temperature values at the inlet and walls, and of the x-derivative at the outlet.

13 Solution visualization

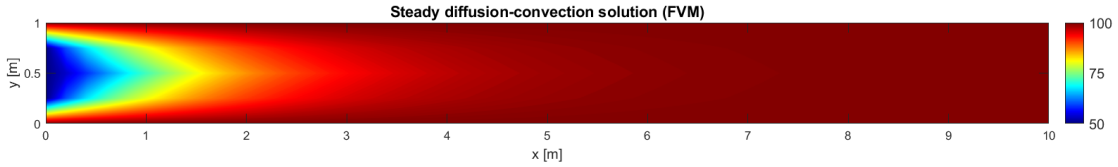
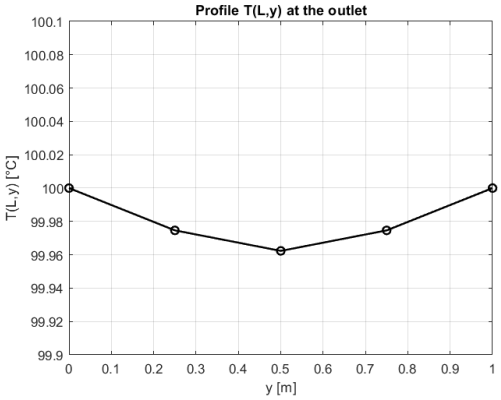
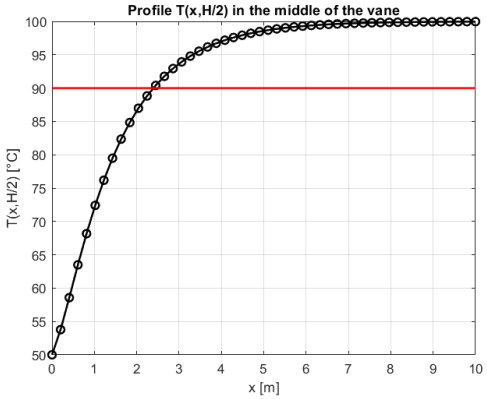


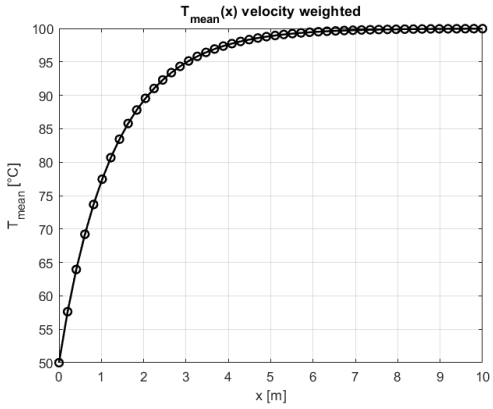
FIGURE 7
Temperature Contour Plot - 2D Solution Plot



(A) Outlet Temperature $T_o(y) = T(L, y)$



(B) Centerline Temperature $T_c(x) = T(x, H/2)$



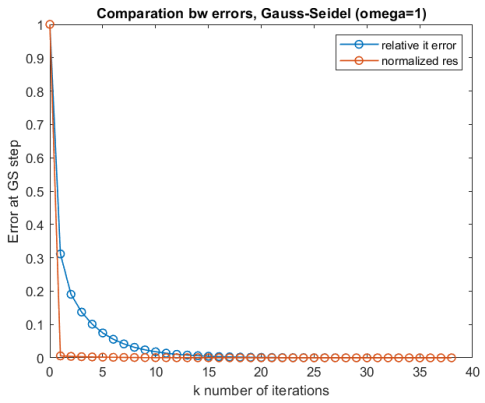
(C) Velocity-weighted Mean Temperature

FIGURE 8
Combined figure showing temperatures and x_e such that $T\left(x_e, \frac{H}{2}\right) = 0.9T_{wall}$

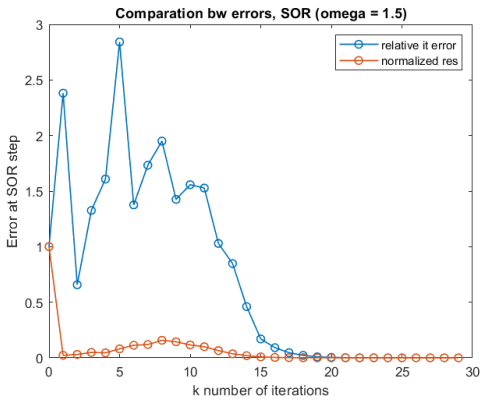
We notice, in figure 8b, that the location x_e such that $T_c(x_e)$ reaches 90% of T_{wall} is approximately $2.4\ m$

14 Iterative methods

In order to compute the solution with the Gauss-Seidel and SOR iterative methods, we started with the initial guess $T_0 = b$, i.e. the source term. Then, we solved the linear system with these methods, with $\omega = 1.5$ for SOR, and plotted the relative iteration error and the normalized residual for each iteration needed for the methods to converge. As we can see in the plots (figures 9a and 9b) both methods converge: GS in 38, SOR in 29 iterations, even if SOR's convergence is not monotone.



(A) Errors computed at Gauss-Seidel steps



(B) Errors computed at SOR steps

FIGURE 9

15 Nusselt number Nu

The Nusselt number Nu is an important dimensionless number in heat transfer problems. It expresses the importance of convection with respect to the conduction in a heat transfer process in a fluid. In other words, it represents the relation between the total heat transfer (convection and conduction) and the purely conductive heat transfer at the fluid-solid boundary. Generally, it is defined as

$$Nu = \frac{q_{conv}}{q_{cond}}.$$

In our case, for laminar flows in plane channels, the Nusselt number $Nu(x)$ decreases in the entrance region and far from the inlet it approaches a constant value independent of the Péclet number. In this case, the Nusselt number is expressed as

$$Nu = \frac{(2H)q}{k\Delta T}$$

where $q = k\partial T/\partial n$ is the heat flux through the wall and $\Delta T = T_{wall} - T_{mean}$ is the difference between the fluid temperature at the wall and the velocity-weighted mean temperature. Therefore, it becomes

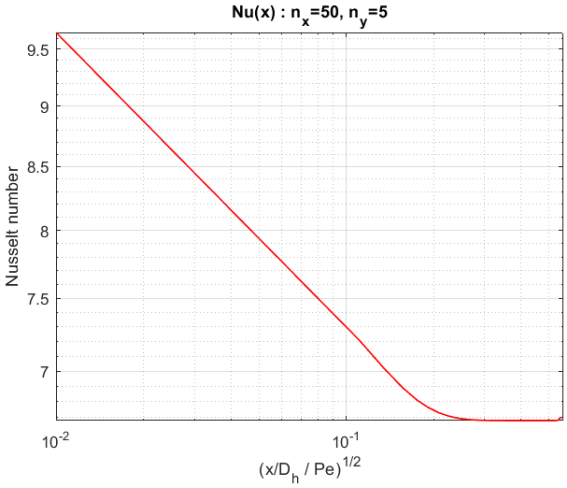
$$Nu = \frac{\frac{\partial T}{\partial n}}{\frac{\Delta T}{2H}}.$$

The Nusselt number for uniform wall temperature $Nu_T(x)$ decreases in the entrance region, and for any Péclet number it converges to 7.54 far enough from the inlet.
To compute $Nu_T(x)$ along the lower wall, we discretized the normal derivative from the fluid to the solid as $\Delta T/\Delta y$, where ΔT is the difference between the wall temperature and the temperature of the upper element close to the wall, leading to the code computation $(T_wall - T(end-1,:))/dy$. In addition, we did not use the u_{mean} value computed before to calculate T_{mean}, but we discretized the integral in the denominator of the equation

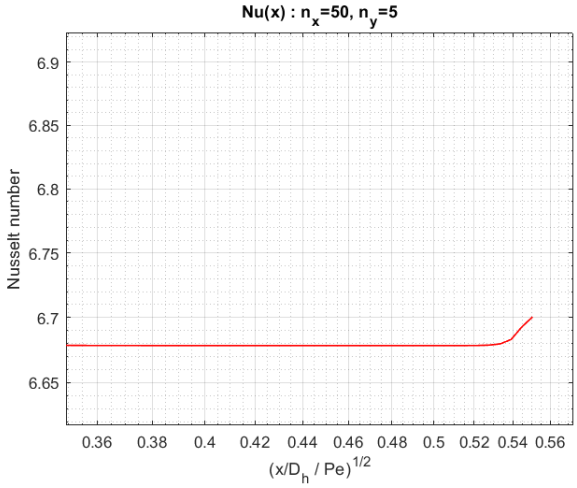
$$T_{mean}(x) = \frac{\int_0^H u_x(y)T(x,y)dy}{\int_0^H u_x(y)dy}.$$

This decision is a consequence of the coarse mesh, which leads to a 10% error between u_{mean} and the numerical results.

FIGURE 10
Nusselt number computed at lower wall.



(A) Nusselt number $Nu_T(x)$ at lower wall



(B) Nusselt number $Nu_T(x)$ at lower wall at the outlet

Some comments on the Nu curve (figure 10a) can be highlighted: first of all, we notice that we obtain a lower magnitude with respect to the reference graph. This is due to the coarse mesh, since the normal derivative at the wall is underestimated because of a large dy with respect to the height. In fact, the convergence value was expected close to 7.54, but instead our curve converges to 6.68. However, the shape meets the expectations. The Nusselt number decreases in the inlet region and then converges as the fluid moves away from the inlet. Looking closer to the outlet values, we notice that there is a slight increase just at the end of the plates. To explain this, we look at the heat equation:

$$\nabla \cdot (\rho \mathbf{u} T) = \nabla \cdot (\Gamma \nabla(T)).$$

The left-hand side is the convection term, and the right-hand side represents the conduction. At the outlet, the imposed boundary condition is the Neumann condition $\partial T/\partial x = 0$. Therefore, if we look at the heat equation at the outlet, the conductive term becomes (assuming Γ constant):

$$\nabla \cdot (\Gamma \nabla(T)) = \nabla \cdot \left(\Gamma \left(\cancel{\frac{\partial T}{\partial x}} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} \right) \right) = \Gamma \frac{\partial^2 T}{\partial y^2}$$

With this boundary condition, the term $\partial T/\partial x$ cancels out of the conduction term. Therefore, recalling the definition of the Nusselt number $Nu = q_{conv}/q_{cond}$, this leads to a decrease in the denominator at the outlet, leading to an increase of the Nusselt number.

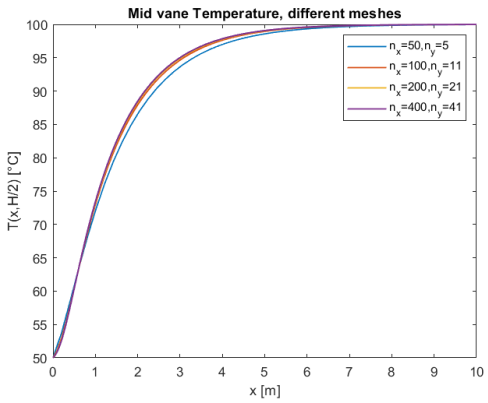
Moreover, the increase in the Nusselt number can also be explained numerically by the fact that the Neumann boundary condition $\partial T/\partial n = 0$ is not perfectly compatible with the actual temperature profile. The discretization forces a slight gradient in the y-direction near the outlet, which artificially increases the temperature difference ΔT, leading to an artificially higher Nusselt number

16 Results with a finer mesh

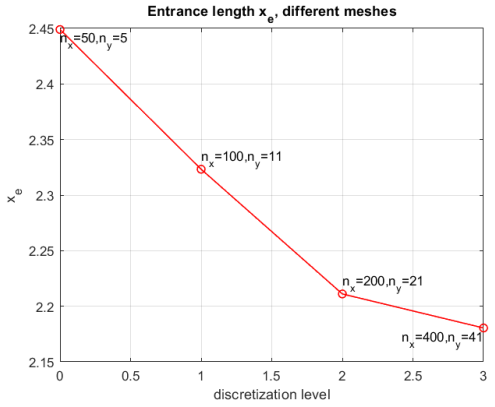
We refine the mesh with a new number of intervals $(n_x, n_y) = (100, 11), (200, 21), (400, 41)$, getting CVs of half the size each step. This leads to a higher computational cost but a more precise discretization, recalling that UD is a first order stable scheme and CD a second order one. This leads to a finer discretization of the mid vane temperature (figure 11a), by getting a more accurate numerical value for the diffusion from the wall to the center of the vane, obtaining an entrance length (figure 11b) closer to the inlet; moreover, we see a change towards $100°C$ for the outlet temperature (figure 11c) due to the better imposition of Neumann BC.

FIGURE 11

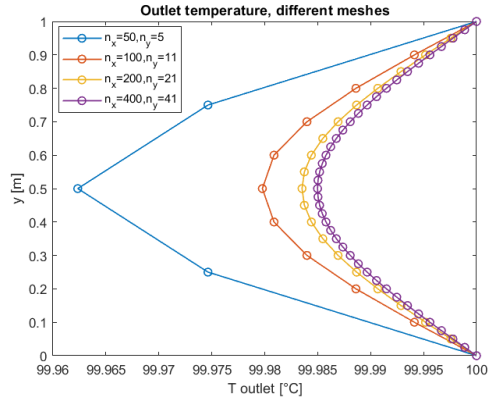
Combined figure showing temperatures and x_e such that $T\left(x_e, \frac{H}{2}\right) = 0.9T_{wall}$



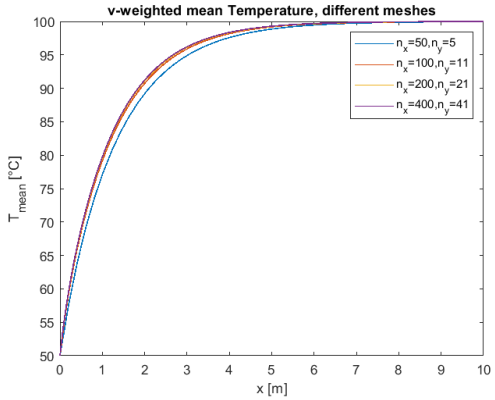
(A) Temperature in the middle of the vane with the different meshes



(B) Corresponding entrance lengths



(C) Temperature at the outlet with the different meshes



(D) Velocity-weighted mean temperature with different meshes

Inspecting the Nusselt number (figure 12), which we still compute by discretizing the integral at the denominator of T_{mean}, we get again better results, thanks to the increasing accuracy of the discretization of $\frac{\partial T}{\partial n}$ at the solid-fluid interface at the wall. With this more correct tuning of the normal derivative, we get shapes that tend to 7.54, as we theoretically would expect, and steeper rises at the outlet, showing better the Neumann BC effect on conduction. In general, having a better approximation of the fluid leads to a better approximation of the different causes of heat conduction.

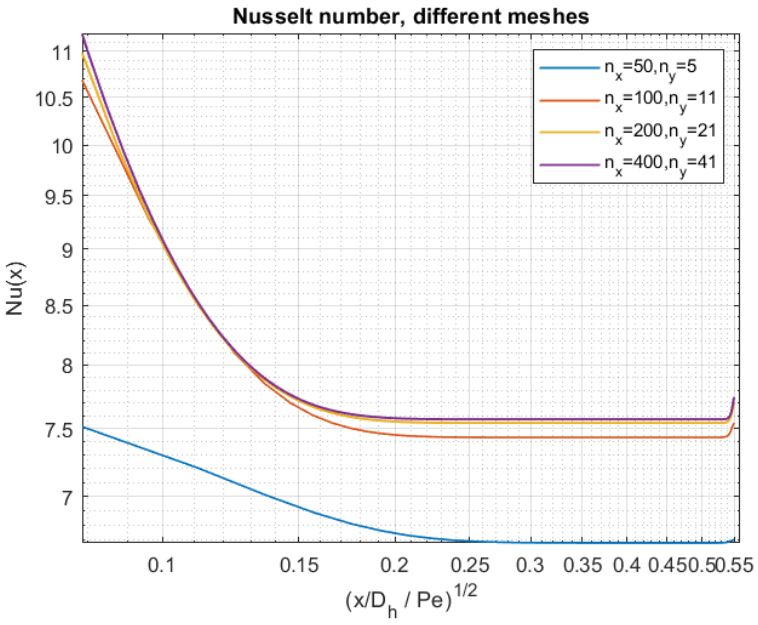


FIGURE 12
Nusselt number with finer meshes

17 QUICK method for convection term

By using the Quadratic upstream interpolation for convective kinetics (QUICK), we guarantee a higher order of convergence, but different errors can still occur; in fact, it could happen that some quantities are underestimated or overestimated due to sharper changes introduced by the double upwind bias; due to the possible unboundedness of the numerical method, caused by the negativity of some coefficients, such as a_{ww}, oscillations in the results can happen too. For better clarity, we show how this new method has been enforced in the code, recalling that for the second column an UD scheme still had to be used:

```
e = ones(N,1);
w_diag = -e*a_w - F_w*0.75 - (1/8)*F_e;
n_diag = -e*a_n;
s_diag = -e*a_s;
e_diag = -e*a_e + (3/8)*F_e;
ww_diag = F_w * (1/8);
p_diag = - w_diag - n_diag - s_diag - ww_diag - e_diag;
b_val = 0;
A_sparse = spdiags([n_diag ww_diag w_diag p_diag e_diag s_diag], [-nx -2 -1 0 +1 +nx], N, N);
sec_column_idx = 2 : nx : (2+nx*(ny-1));
A = full(A_sparse);
for ii=1:ny
    if(sec_column_idx(ii)-2 > 0)
        A(sec_column_idx(ii),sec_column_idx(ii)-2) = 0;
    end
    A(sec_column_idx(ii),sec_column_idx(ii)) = a_p + F_e(sec_column_idx(ii));
    A(sec_column_idx(ii),sec_column_idx(ii)-1) = -a_w - F_w(sec_column_idx(ii));
    A(sec_column_idx(ii),sec_column_idx(ii)+1) = -a_e;
end
b = e*b_val;
```

LISTING 7
Matrix for inner CVs with QUICK before enforcing BCs

As we can see from the different values of the entrance length x_e (figure 13b), with a coarse mesh, x_e is identical to the one obtained with UD. By refining the mesh, thanks to the 3rd order convergence of QUICK for convection, the change towards the real value is faster. This is the same reason why the relative variance of x_e (figure 13d) is initially much steeper than after two refinements: the numerical convection has already converged.

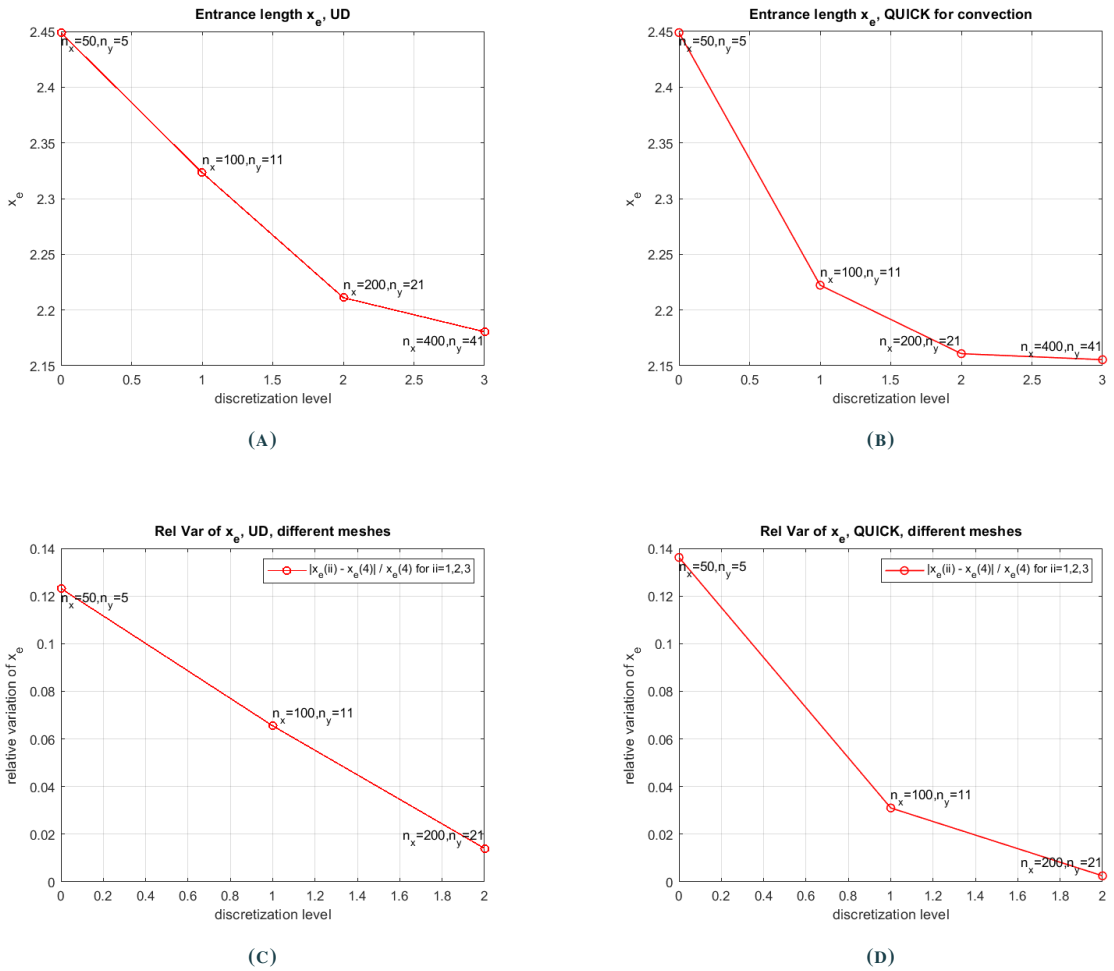


FIGURE 13

18 Different mesh ratios

Different mesh sizes could lead to more efficient results. To visualize the accuracy of different mesh sizes, we compute the minimum value of the Nusselt number for the mesh sizes $(n_x, n_y) = (400, 5), (400, 11), (400, 21)$ with n_x constant and $(n_x, n_y) = (100, 41), (200, 41), (400, 41)$ with n_y constant.

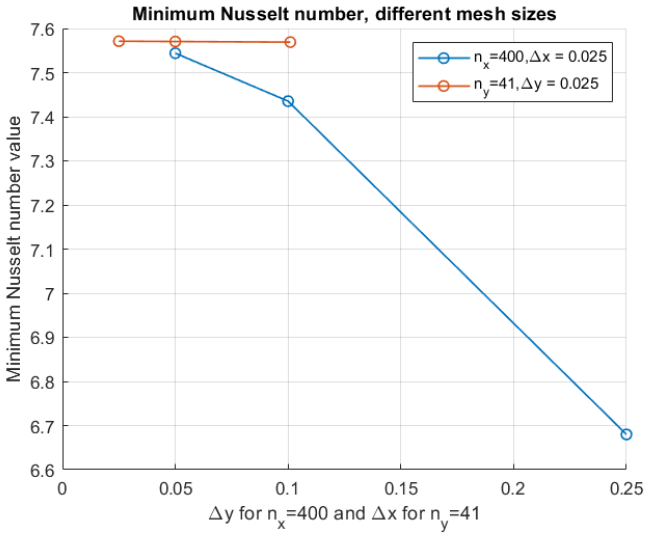
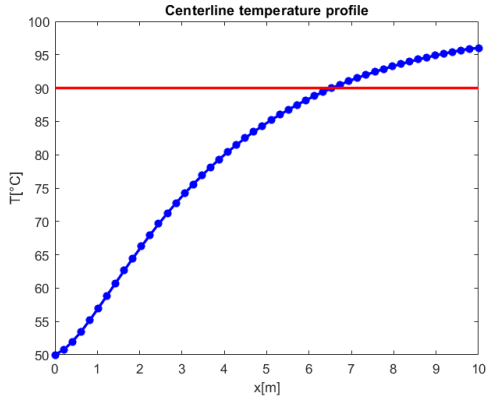


FIGURE 14
Minimum Nusselt number value for different mesh sizes

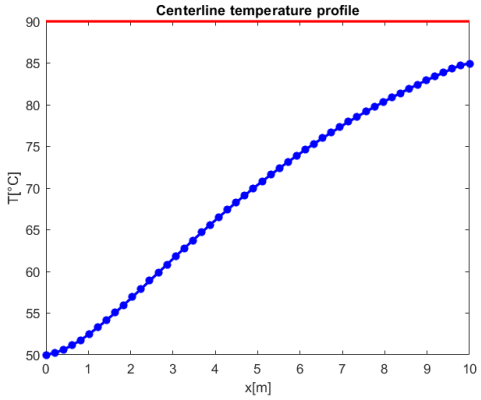
The illustration of the minimum Nusselt number value curves (figure 14) shows important results: first of all, we notice that the Nusselt numbers for meshes with $n_y = 41$ are really close, which suggests that the variation of Δx does not have a relevant impact on accuracy. Moreover, a high number of discretization of the vertical axis leads to accurate computations, since the minimum Nusselt number is close to the expected convergence value of 7.54 (for $n_x = 100, 200, 400$ we obtain $Nu_{min} = 7.5691, 7.5705, 7.5712$, respectively).
Instead, for meshes with fixed $n_x = 400$, we notice an important increase in accuracy as n_y increases (and therefore Δy decreases). For a larger mesh size, the Nusselt number is very far from the convergence value, and as n_y increases, we obtain more accurate results since the Nusselt number approaches 7.54 (for the mesh size $(400, 21)$ $Nu_{min} = 7.5440$).
We can conclude that the size of the vertical axis mesh has a huge influence on results compared to the horizontal discretization. This is because the variation in temperature along the y-axis is larger than the one along the x-axis. Moreover, for cases where n_y is constant, we notice that with fewer mesh elements, hence a lower cost, we obtain approximately the same result as more costly discretizations.

19 Effect of high Péclet numbers

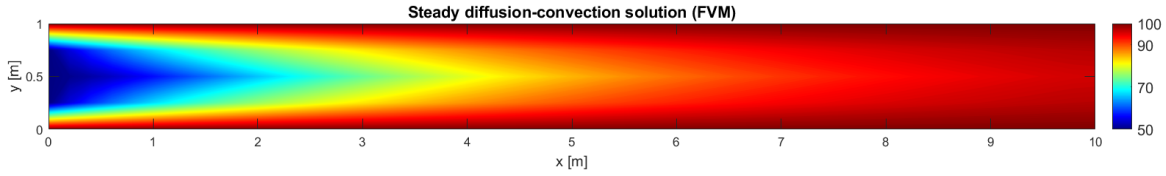
We inspected how the same numerical scheme would have performed with higher Péclet numbers, precisely $Pe = 50,\ 100$. As we can see in figures 15a and 15b, a higher dominance of convection leads to a slower convergence of the temperature to the temperature diffusing from the walls. Despite this, the solution is stable in our context, likely because the scheme we are using (UD) is inherently diffusive and does not produce negative coefficients. Nevertheless, the UD scheme has a drawback: a non-negligible numerical diffusion arises. This phenomenon could be partially mitigated using finer meshes, as shown by comparing contour plots in 15 and 16: in the coarser mesh, the overestimated diffusion is clearly visible starting from the inlet and along the flow direction.



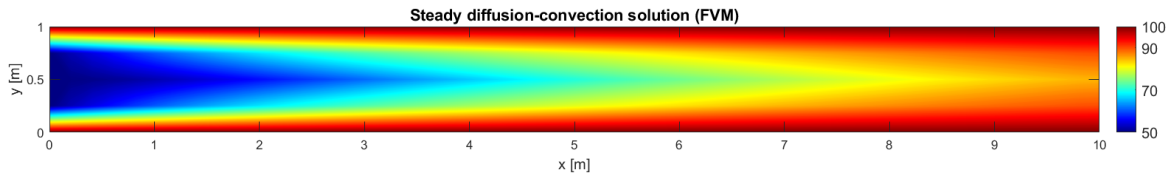
(A) Temperature in the middle of the vane with $Pe = 50, n_x = 50, n_y = 5$



(B) Temperature in the middle of the vane with $Pe = 100, n_x = 50, n_y = 5$

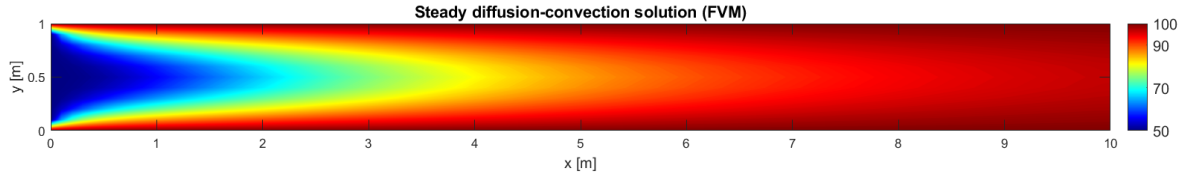


(C) Contour with $Pe = 50, n_x = 50, n_y = 5$

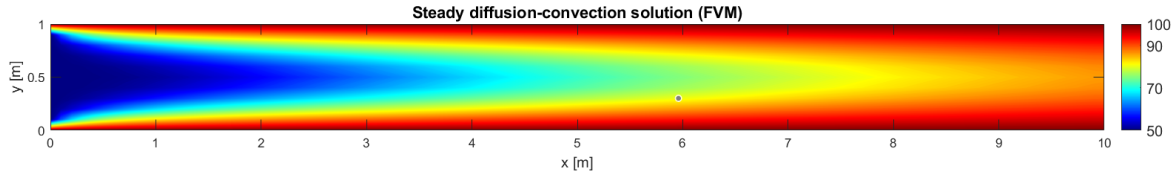


(D) Contour with $Pe = 100, n_x = 50, n_y = 5$

FIGURE 15
$T\left(x, \frac{H}{2}\right)$ and countour plots for $Pe = 50$ and $Pe = 100$



(A) Contour with $Pe = 50, n_x = 100, n_y = 11$



(B) Contour with $Pe = 100, n_x = 100, n_y = 11$

FIGURE 16
Contour plots for $Pe = 50$ and $Pe = 100$ with a finer mesh

Furthermore, we could add that higher Péclet numbers could create some problems of transportiveness, particularly on coarser meshes, due to the possible asymmetry of the behavior among neighboring nodes: for this reason it is important to discretize convection-diffusion equations using a scheme transportive by construction, as UD for example. Moreover, high values of this number can cause oscillations or instability with discretization schemes in which coefficients can become negative (e.g. CD or QUICK).

20 Wall Neumann Boundary conditions

From now on, we set

$$k\frac{\partial T}{\partial n} = q_{wall}$$

to be satisfied on both walls.

20.1 Upper wall discretization

In the upper wall, $\mathbf{n} = (0, +1)$, i.e. north direction. For this reason, we discretize the BC as

$$\frac{\partial T}{\partial n} \approx \frac{T_N - T_P}{\delta y_{PN}}$$

and obtain

$$D_n(T_N - T_P)\Delta x \approx \frac{\Gamma_n}{k} q_{wall}\Delta x$$

, which we substitute in the original algebraic equation (5), but changing Δy to $\frac{\Delta y}{2}$. We finally obtain, for non-corner CVs:

$$a_P T_P = a_E T_E + a_W T_W + a_S T_S + \frac{\Gamma_n}{k} q_{wall}\Delta x$$

where coefficients a_i are the same as in (6) but with Δy changed with $\frac{\Delta y}{2}$. The upper right corner needs to be discretized separately, to force both the Neumann BCs on the wall and on the outlet. Applying the same process as (8), we get:

$$\left(\frac{F_e}{2} + D_w\frac{\Delta y}{2} + D_s\frac{\Delta x}{2}\right) T_P = 0.T_E + \left(\frac{F_w}{2} + D_w\frac{\Delta y}{2}\right) T_W + D_s\frac{\Delta x}{2} T_S + \frac{\Gamma_n}{2k} q_{wall}\Delta x \tag{7}$$

The upper left corner will be considered part of the inlet and treated as in 7.

20.2 Lower wall discretization

In the lower wall, $\mathbf{n} = (0, -1)$, i.e. the south direction. For this reason, we discretize the Neumann BC at the lower wall as

$$\frac{\partial T}{\partial n} \approx \frac{T_S - T_P}{\delta y_{SP}}$$

and we obtain

$$D_s(T_S - T_P)\Delta x \approx \frac{\Gamma_s}{k} q_{wall} \Delta x$$

which we substitute in the original algebraic equation (5), but changing Δy to $\frac{\Delta y}{2}$. We finally obtain, for non-corner CVs:

$$a_P T_P = a_E T_E + a_W T_W + a_N T_N + \frac{\Gamma_s}{k} q_{wall} \Delta x$$

where coefficients a_i are the same as in (6) but with Δy changed with $\frac{\Delta y}{2}$. The lower right corner needs to be discretized separately, to force both the Neumann BCs on the wall and on the outlet. Applying the same process as (8), we get:

$$\left(\frac{F_e}{2} + D_w \frac{\Delta y}{2} + D_n \frac{\Delta x}{2}\right) T_P = 0.T_E + \left(\frac{F_w}{2} + D_w \frac{\Delta y}{2}\right) T_W + D_n \frac{\Delta x}{2} T_N + \frac{\Gamma_s}{2k} q_{wall} \Delta x \tag{8}$$

Again, the lower left corner will be considered as part of the inlet.

20.3 Physical consistency check

As shown in the two plots below, the solution is physically reasonable since the temperature increases uniformly along the x direction, consistent with the constant Neumann boundary condition at the walls. Moreover, with regard to the y direction, the solution is high in the vicinity of the walls and decreases as we approach the centerline, where it attains its minimum. This behavior results from heat diffusion from the walls toward the bulk flow.

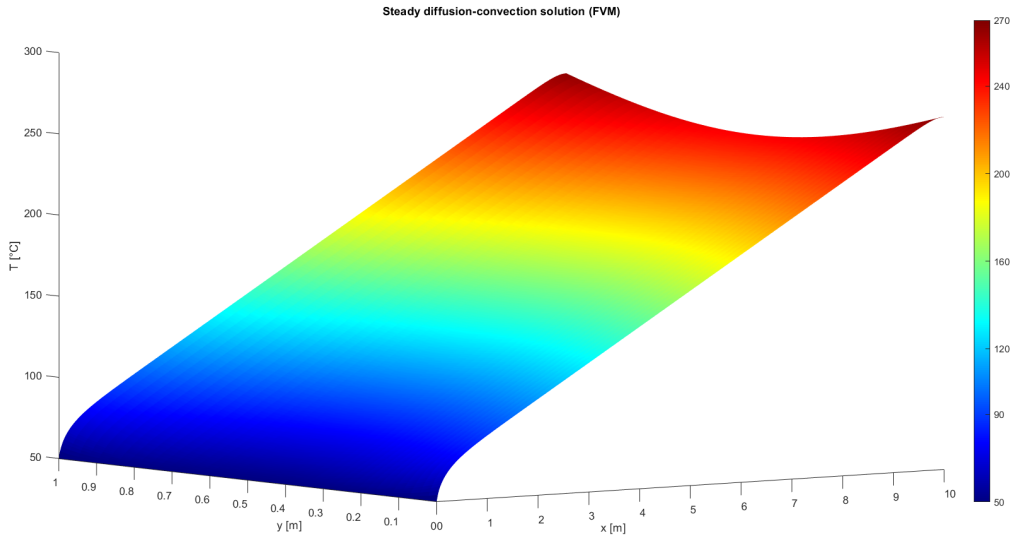


FIGURE 17
Surf plot with $Pe = 16.5, n_x = 400, n_y = 41$

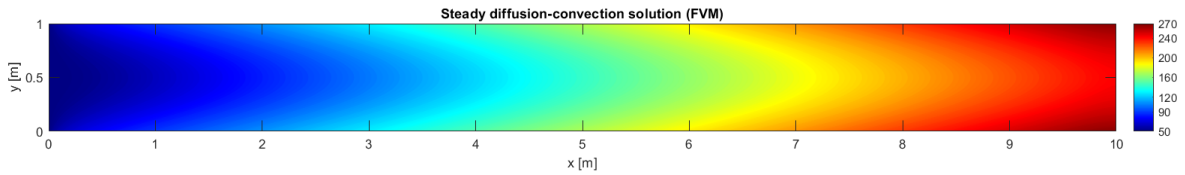


FIGURE 18
Contour plot with $Pe = 16.5, n_x = 400, n_y = 41$

20.4 Boundary conditions plots

After setting $q_{wall} = 10\ W/m^2$, we check the quality of the boundary conditions implemented:

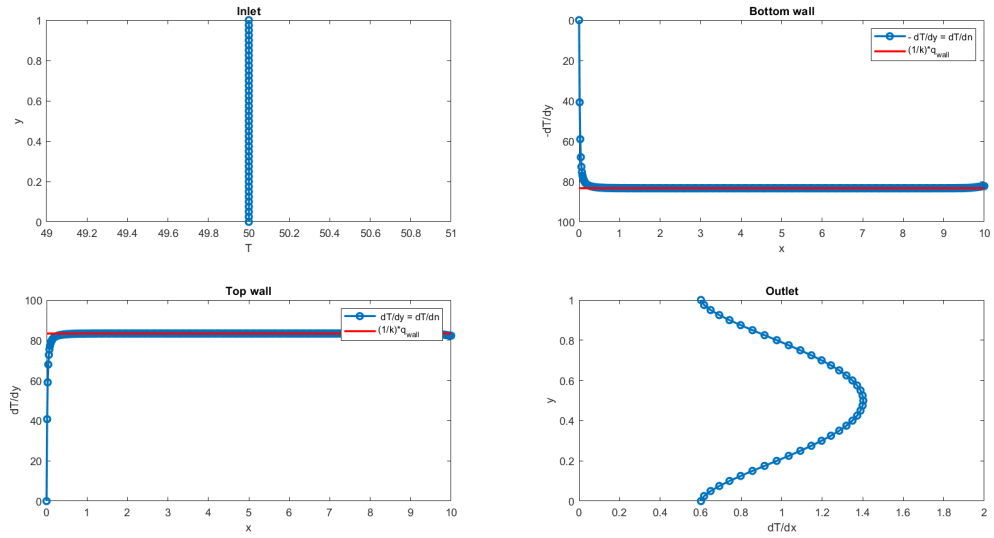


FIGURE 19
Boundary conditions with $Pe = 16.5, n_x = 400, n_y = 41$

Due to the high temperature reached near the outlet at $Pe = 16.5$, it is difficult to accurately discretize the normal derivative in the eastward direction, especially on a coarse mesh, since the temperature gradient remains steep.

For a much higher Péclet number, for instance $Pe = 1650$, it could actually become easier to discretize this Neumann boundary condition, since the temperature gradient is lower due to the convection effect that keeps the fluid colder alongside the domain. As shown in section 19, our numerical solution would not even reach $0.9T_{\text{wall}}$ in the centerline, which ensures a smoother numerical enforcement of the condition $\dfrac{\partial T}{\partial x} = 0$.

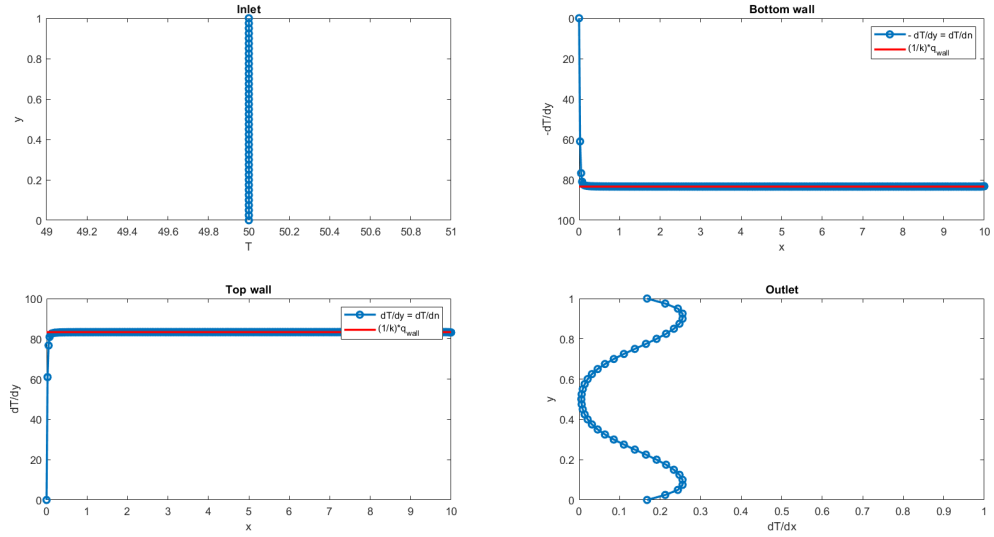


FIGURE 20
Boundary conditions with $Pe = 1650$, $n_x = 400$, $n_y = 41$.

20.5 Nusselt number

Examining the discretization quality of the local Nusselt number $Nu_q(x)$ along the lower wall, we observe that with a grid of $n_x = 400$, $n_y = 41$, the computed value approaches the asymptotic limit of approximately 8.235, which is consistent with the theoretical value expected for a uniform heat flux boundary condition.

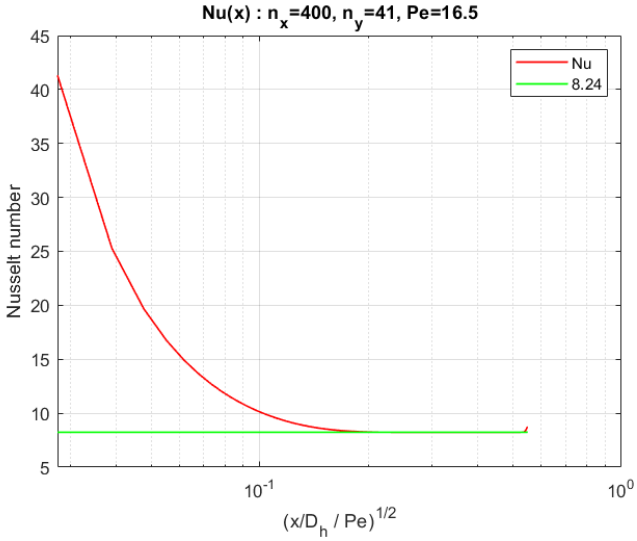


FIGURE 21
Nusselt number with $Pe = 16.5, n_x = 400, n_y = 41$

21 Global energy balance

To verify if the thermal energy is globally conserved, we perform a global heat balance. The general formula is

$$\int_{\partial\Omega} (k\nabla T - \rho c_p \mathbf{u} T) \cdot \mathbf{n} dS = 0$$

In this case, we have four boundaries (top, bottom, inlet, outlet), therefore, we can write the integral as a sum of the corresponding integral for each boundary.

$$\int_0^L (k\partial_y T)\Big|_{y=H} dx - \int_0^L (k\partial_y T)\Big|_{y=0} dx + \int_0^H (k\partial_x T - \rho c_p u_x T)\big|_{x=L} dy - \int_0^H (k\partial_x T - \rho c_p u_x T)\big|_{x=0} dy = 0$$

For the top and bottom plates, we consider only the conductive term, since the vertical velocity component u_y is equal to zero. Instead, for the inlet and outlet, we have to consider the convective heat transfer, generated by the horizontal velocity component u_x, and the conductive term.
Numerically, we first compute the flux density q for the four boundaries with

$$q = k\frac{\partial T}{\partial n}$$

The derivatives are calculated in the same way as in 15, therefore, we consider the normal derivative approximately $\Delta T/dn$, where ΔT is the difference between the boundary temperature and the first close fluid temperature. Afterward, we compute the total heat flux for each boundary by summing the flux density for each node and multiplying by the elementary length. Then, we compute the convective terms for the inlet and outlet boundaries and add them to the conductive terms. Finally, we perform the balance by summing up all the terms.

```
q_top_wall = k*((T_mat(N_y, :) - T_mat(N_y-1, :))/dy);
q_bottom_wall = k*((T_mat(1, :) - T_mat(2, :))/dy);
q_inlet = k*((T(:, 1) - T(:, 2))/ dx);
q_outlet = k*((T(:,end) - T(:,end-1)) ./ dx);

Q_top_wall = sum(q_top_wall) * dx;
Q_bottom_wall = sum(q_bottom_wall) * dx;
Q_inlet = sum(q_inlet) * dy;
Q_outlet = sum(q_outlet) * dy;

flux_inlet = rho*c_p*sum(ux'.*T(:,1))*dy;
```

```
flux_outlet =  -rho*c_p*sum(ux'.*T(:,N_x))*dy;

inlet = flux_inlet + Q_inlet;
outlet =  flux_outlet + Q_outlet;
upper_wall = Q_top_wall;
lower_wall = Q_bottom_wall;

balance = upper_wall + lower_wall + inlet + outlet;
```

LISTING 8
Numerical computation heat fluxes

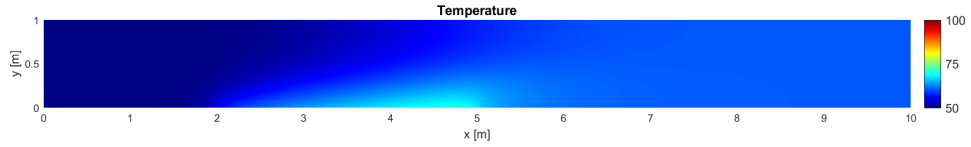
We remark that in the MATLAB code the flux densities q for the bottom wall and the inlet have opposite signs with respect to the equation 21 since there the normal vector is opposite to the y and x directions, respectively. With this computation, we obtain the following values: $Q_{top} = 99.7376\,W/m$, $Q_{bottom} = 99.7376\,W/m$, $Q_{inlet} = 45.326\,W/m$, $Q_{outlet} = -245.392\,W/m$. And for the balance, we obtain $balance = -0.5907\,W/m$.
First, we notice that the flux values for the top and bottom plates and for the inlet are positive, meaning that the flux enters the fluid. In fact, the top and bottom plates have an imposed positive flux that enters the system; therefore, the fluid heats up. At the inlet, although the fluid enters at a low temperature, the convective term of the heat flux is much higher than the conductive term, leading to a positive heat flux. For the same reason, at the outlet, the heat flux is negative.
Performing the balance, we obtain a value of around $-0.6\,W/m$, which is close to zero. This is reasonable since there is no source of loss in the system, and therefore the energy is conserved.
However, if we perform the balance considering the imposed heat flux Neumann condition $k\partial T/\partial n = q_{wall}$ at the top and bottom walls, we can rewrite the equation as follows:

$$\int_0^L q_{wall}dx + \int_0^L q_{wall}dx + \int_0^H (k\partial_x T - \rho c_p u_x T)\big|_{x=L} dy - \int_0^H (k\partial_x T - \rho c_p u_x T)\big|_{x=0} dy = 0$$
$$\Rightarrow q_{wall}L + q_{wall}L + \int_0^H (k\partial_x T - \rho c_p u_x T)\big|_{x=L} dy - \int_0^H (k\partial_x T - \rho c_p u_x T)\big|_{x=0} dy = 0$$

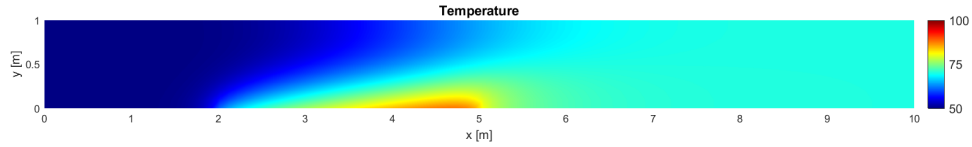
With this equation, we obtain a balance result of $-0.0660\,W/m$, which is much better compared to the one obtained by the numerical simulation. In fact, we notice a small difference in the heat flux of the plates, which in this case are $Q^*_{top} = Q^*_{bottom} = 100.0\,W/m$, and with the numerical simulation we obtain $Q_{top} = Q_{bottom} = 99.7376\,W/m$. This error of 0.26% on the flux is due to the approximations performed during the simulation, and it is stood up in the balance, since it is taken into account twice (in the top and bottom plate heat flux).
Therefore, we can conclude that the numerical simulation is accurate.

22 System with non-uniform wall boundary conditions

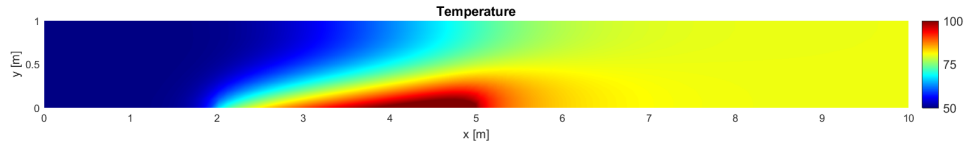
In this last part, we have non-uniform wall boundary conditions. A positive heat flux is located in the region $2 \leq x \leq 5$ of the lower wall. Everywhere else, walls are considered insulated/adiabatic ($k\partial T/\partial n = 0$).
With these boundary conditions, if we want the outlet temperature to be $60°C$, $70°C$, or $80°C$, we found with an empirical approach, in particular checking the outlet temperature plots for different heat flux values, that we need approximately heat fluxes $q_{wall} = [3.4,\ 6.5,\ 9.8]\ W/m^2$, respectively. Plotting an approximation of the outlet temperature we got versus the heat flux, we can observe that there is a linear relation between them, as shown in figure 24.



(A) Temperature profile for $q_{wall} = 3.4\text{W/m}^2$

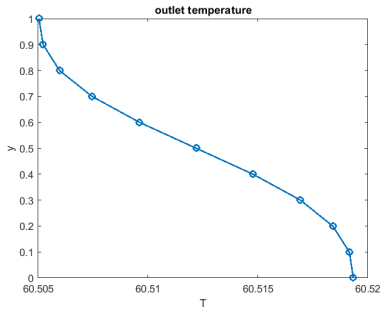


(B) Temperature profile for $q_{wall} = 6.5\text{W/m}^2$

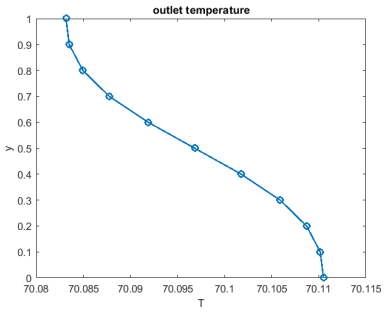


(C) Temperature profile for $q_{wall} = 9.8\text{W/m}^2$

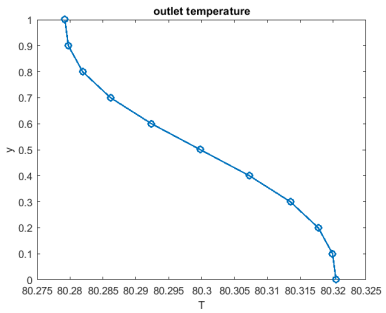
FIGURE 22
The simulations have been performed setting $n_x = 100$ and $n_y = 11$



(A) Outlet temperature, $q_{wall} = 3.4\text{W/m}^2$



(B) Outlet temperature, $q_{wall} = 6.5\text{W/m}^2$



(C) Outlet temperature, $q_{wall} = 9.8\text{W/m}^2$

FIGURE 23
Outlet temperature for non-uniform wall boundary conditions

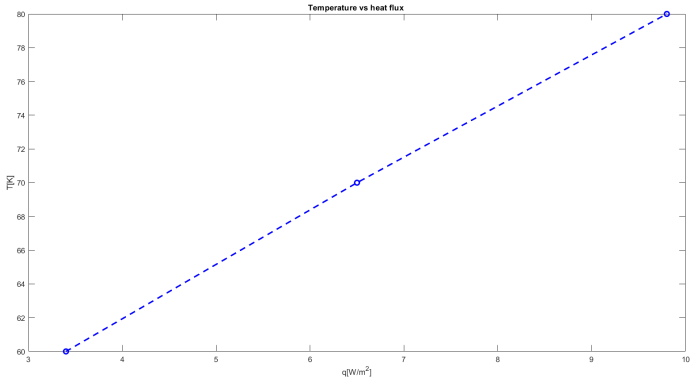


FIGURE 24
Outlet temperature versus heat flux

The temperature variation (figure 22) between the two plates has a similar shape between the three cases. In the images, we can see that the fluid enters at a cold temperature of 50°C, and starting from $x = 2$ m close to the lower plate, it starts to heat up. Between $x = 2$ m and $x = 5$ m, there is a region in the lower part, where the fluid temperature is at a maximum, and this is due to the presence of the heat flux q_{wall}. The fluid does not heat up homogeneously: the heat comes from the bottom plate and propagates upward, leading to a progressive increase in temperature. Once $x = 5$ m is passed, the fluid reaches an almost constant temperature throughout the cross-section, since the walls are considered adiabatic. The plots in figure 23 confirm that with the relative heat flux q_{wall}, we obtain the desired outlet temperatures, which can be considered approximately constant in the cross-section.